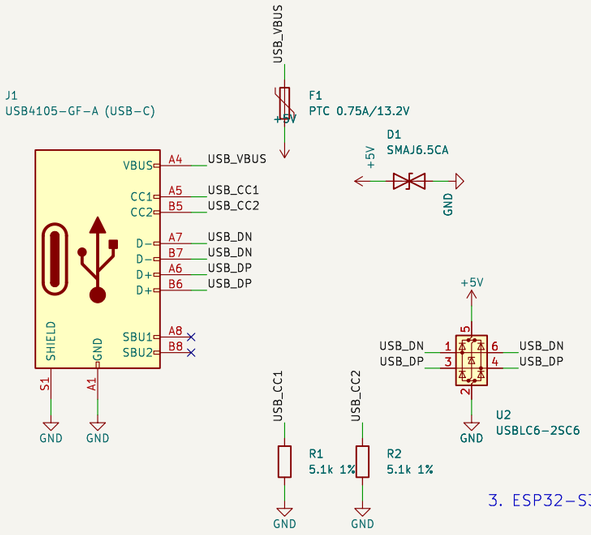
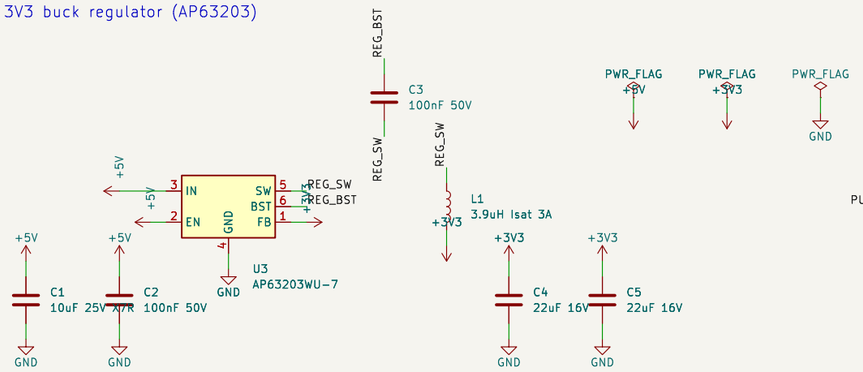
1. USB-C input, PTC fuse, TVS, USB ESD

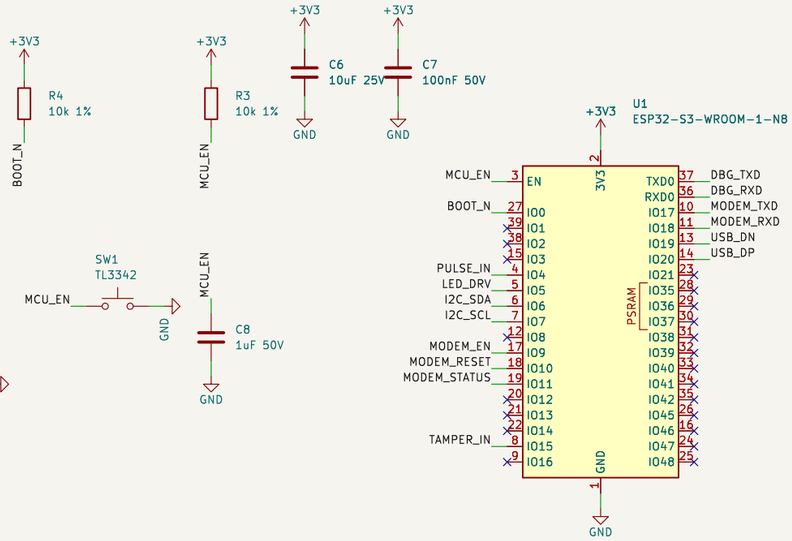


2. 3V3 buck regulator (AP63203)

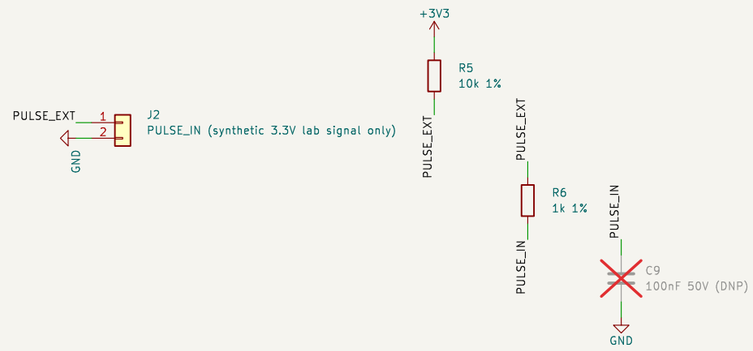


Power path: USB_VBUS -> F1 (PTC) -> +5V -> U3 AP63203 -> L1 -> +3V3. PWR_FLAGs mark rails fed through passives for ERC.

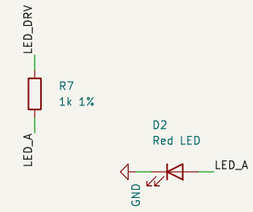
3. ESP32-S3-WROOM-1-N8, EN RC, RESET/BOOT



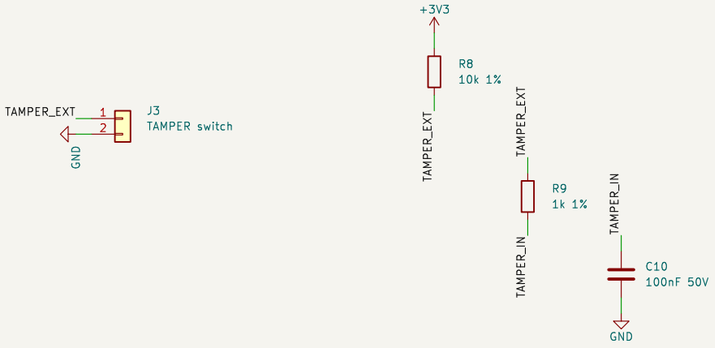
4. Synthetic pulse input — 3.3V LAB SIGNAL ONLY



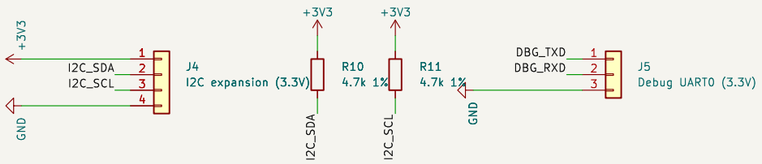
5. Status LED



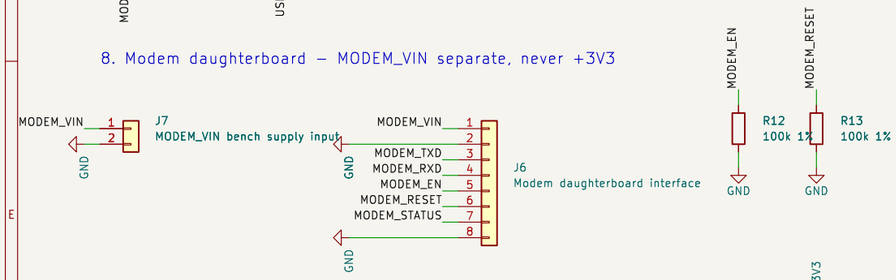
6. Enclosure tamper switch (advisory only)



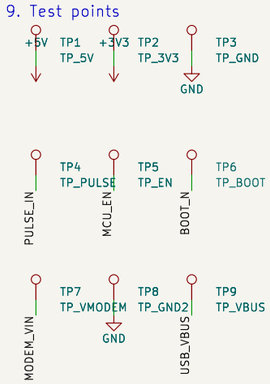
7. I2C expansion and debug UART



8. Modem daughterboard — MODEM_VIN separate, never +3V3



9. Test points



SCHEMATIC FREEZE CANDIDATE rc1 — NOT FABRICATION READY. Generated from design_v02.py.
Bench-only: current-limited low-voltage supply + synthetic 3.3V pulse source.
Never connect to a live gas meter, gas line, valve, mains or field wiring.
Exact parts selected (see PART_SELECTION.md); PCB layout and bench bring-up pending.

Smart Gas Meter IoT — lab prototype

Sheet: /
File: smart_gas_meter_iot.kicad_sch

Title: Smart Gas Meter IoT — carrier board v0.2 (controlled rebuild)

Size: A3	Date: 2026-10-24	**Rev: 0.2-rc1**
KiCad E.D.A. 9.0.3		Id: 1/1